

Atomistic Simulation of HF Etching Process of Amorphous Si_3N_4 Using Machine Learning Potential

Changho Hong,^{||} Sangmin Oh,^{||} Hyungmin An, Purun-hanul Kim, Yaeji Kim, Jae-hyeon Ko, Jiwoong Sue, Dongyeon Oh, Sungkye Park, and Seungwu Han*



Cite This: *ACS Appl. Mater. Interfaces* 2024, 16, 48457–48469



Read Online

ACCESS |



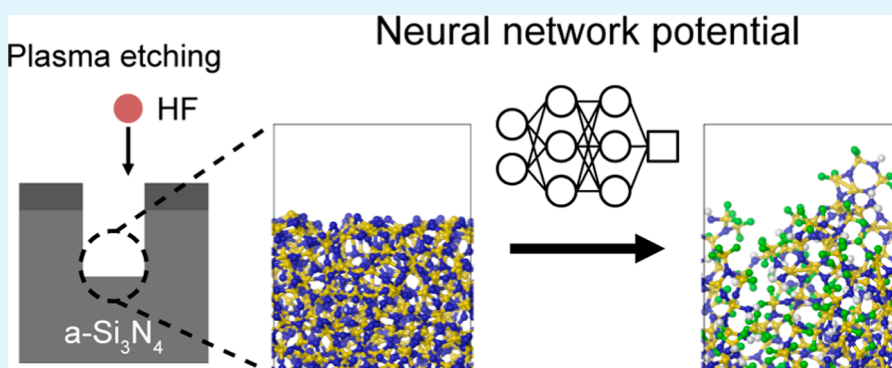
Metrics & More



Article Recommendations



Supporting Information



ABSTRACT: An atomistic understanding of dry-etching processes with reactive molecules is crucial for achieving geometric integrity in highly scaled semiconductor devices. Molecular dynamics (MD) simulations are instrumental, but the lack of reliable force fields hinders the widespread use of MD in etching simulations. In this work, we develop an accurate neural network potential (NNP) for simulating the etching process of amorphous Si_3N_4 with HF molecules. The surface reactions in diverse local environments are considered by incorporating several types of training sets: baseline structures, reaction-specific data, and general-purpose training sets. Furthermore, the NNP is refined through iterative comparisons with the density functional theory results. Using the trained NNP, we carry out etching simulations, which allow for detailed observation and analysis of key processes such as preferential sputtering, surface modification, etching yield, threshold energy, and the distribution of etching products. Additionally, we develop a simple continuum model, built from the MD simulation results, which effectively reproduces the surface composition obtained with MD simulations. By establishing a computational framework for atomistic etching simulation and scale bridging, this work will pave the way for more accurate and efficient design of etching processes in the semiconductor industry, enhancing device performance and manufacturing precision.

KEYWORDS: machine learning potential, plasma etching, molecular dynamics, density functional theory, multiscale simulation, silicon nitride

1. INTRODUCTION

Modern semiconductor devices consist of complex three-dimensional structures with critical dimensions spanning only a few nanometers. An example is the high-density flash memory used in many electronic devices. To increase storage density, multilayer deep vertical structures are fabricated with high aspect ratios of ~ 100 and hole radii of 100 nm.^{1,2} Achieving geometric integrity of such highly anisotropic structures requires sophisticated dry-etching processes using plasma gases.^{3–6} In optimizing the process conditions and improving yields, the computational techniques have been instrumental, playing a pivotal role in advancing process technology.^{7,8} For instance, the composition and spatial distribution of plasma gases are simulated using computational fluid dynamics (CFD) coupled with chemical kinetics and equilibrium theory.⁹ The

simulation results are useful in maintaining the structural integrity across the wafer, for instance.¹⁰

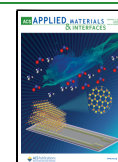
As flash memory is further scaled by increasing the number of stacking layers and reducing the pitch dimension, the demand for accurate prediction of the etch profile is increasing to minimize shape distortions.¹¹ This, in turn, necessitates a simulation module for describing plasma–surface interactions. A popular approach is the plasma-chemistry-based Monte

Received: May 14, 2024

Revised: July 19, 2024

Accepted: August 21, 2024

Published: August 28, 2024



Carlo method.¹² In this mesoscale simulation, the material is coarse-grained into voxels with side lengths of 5–10 nm, and possible chemical interactions are defined with reaction rates. For each incident plasma particle, reactions such as etching, deposition, and passivation are determined probabilistically by the Monte Carlo method, and the state of the voxels evolves accordingly. (It is also worth noting a development using the level set method.¹³) In tandem with the CFD simulations yielding distributions for ions and neutrals near surface, the Monte Carlo method selects reactions and products based on the composition of the voxel and the incident particle, thereby emulating etch profiles depending on the process conditions. However, the model includes many parameters for surface reactions, most of which are difficult to determine precisely from the known chemistry. The experimental information is not sufficiently detailed to circumvent this problem, either. Furthermore, as new etching gases are introduced, the phenomenological fitting of simulation parameters is increasingly difficult.⁸

The foregoing discussions highlight the limitation of current computational framework in dry-etching simulation with plasma, that mainly originates in the lack of detailed atomistic information on the plasma–surface interaction. This can be augmented by bottom-up approaches based on molecular dynamics (MD) simulations that directly emulate atomic processes in ion etching. By following the atomic trajectory using Newton's law of motion, the MD simulation can enlighten the etching mechanism and offer various parameters such as etching yield without resorting to empirical information. For instance, physical sputtering by inert gas and reactive ion etching (RIE) has been studied using classical potentials or reactive force-fields (ReaxFFs).^{7,14–18} From the simulation, useful information such as etch rate was obtained together with fundamental understanding on the etching process. However, the available potential model is highly limited, and parametrization requires human expert knowledge and a long period of development.¹⁹ As an alternative, the density functional theory (DFT) is a robust and accurate approach with little empiricism. However, performing MD for etching simulations using DFT, which spans multiple nanoseconds with at least thousands of atoms, is intractable.

Recently, machine learning potentials (MLPs) have gained significant attention as they can evaluate energies with accuracy comparable to DFT, yet at a significantly lower cost.²⁰ This efficiency enabled the accelerated prediction of crystal structures,²¹ phase diagrams,²² and thermal and diffusional properties in bulk materials.^{23,24} MLPs have also proven to be useful in various surface simulations, such as catalysis at surfaces,²⁵ physical sputtering simulations,²⁶ and surface oxidation processes.²⁷ While the use of MLPs in atomistic etching simulations seems promising, preparing the DFT training data presents significant challenges.²⁸ This is because sampling chemical reactions for these simulations requires long-term MD simulations, which are costly with DFT. Furthermore, it is essential to accurately capture the diverse environmental conditions encountered during simulations.

Motivated by foregoing discussions, in this study, we suggest a protocol that can produce reliable MLPs for surface reactions, which can be used in atomistic etching simulation. This protocol consists of several types of training set; baseline structures, reaction-specific data, together with general-purpose training sets. Furthermore, the MLP is refined through iterative comparisons with DFT results. As a showcase example, we

simulate the RIE of amorphous Si₃N₄ (a-Si₃N₄) by HF molecules using MD simulations with a neural network potential (NNP). In industry, hydrofluorocarbons are more favored when dry etching silicon nitride,²⁹ but for simplicity we focus on HF gas in this work. To examine the kinetic effect of reactive ions, HF molecules are incident with kinetic energy of 10–50 eV, which is the typical energy of ions colliding with the surface without bias.³⁰ We conduct MD simulations of a-Si₃N₄ being etched by HF molecules and analyze etching characteristics, including the etching yield and surface profile, with respect to the energy and angle of incidence. Furthermore, we develop a simple mesoscale model, built from our MD simulation results, which effectively reproduces the surface composition obtained with MD simulations.

The rest of the paper is organized as follows: in Section 2, we explain the types of NNP employed in this work and training method (Section 2.1), DFT parameters for generating the training data (Section 2.2), and the simulation setup for etching process (Section 2.3). In Section 3.1, we focus on the training of the NNP, including data set construction and performance evaluation. In Section 3.2, we examine the etching process under various incident energies and angles using the trained NNP. The microscale simulation results are connected to a mesoscale model in Section 3.3. Finally, we summarize and conclude this work in Section 4.

2. COMPUTATIONAL METHODS

2.1. Hybrid Neural Network Potential. During RIE, atoms experience close interactions due to high-energy impacts. Sampling these configurations using DFT would require significant computational resources. As an alternative, these close atomic interactions can be effectively modeled with the screened Coulomb potential, which integrates well with MLPs.^{31–33} Following prior studies, we employ the pairwise Ziegler–Biersack–Littmark (ZBL)³⁴ potential to simulate the interactions between energetic atoms. Thus, the potential energy (E_{pot}) is decomposed as follows

$$E_{\text{pot}} = E_{\text{ZBL}} + E_{\text{NNP}} \quad (1)$$

where E_{ZBL} and E_{NNP} represent the energy from ZBL potential and NNP, respectively. In eq 1, E_{ZBL} is defined as the sum of pairwise repulsive potentials E_{ij}^{ZBL}

$$E_{\text{ZBL}} = \sum_{i=1}^{N_a} \sum_{j \in \mathcal{N}(i)} \frac{1}{2} E_{ij}^{\text{ZBL}} \quad (2)$$

$$E_{ij}^{\text{ZBL}} = \frac{1}{4\pi\epsilon_0} \frac{Z_i Z_j e^2}{r_{ij}} \Phi(r_{ij}/a) + S(r_{ij}) \quad (3)$$

where N_a is the number of atoms in a structure, $\mathcal{N}(i)$ denotes the neighborhood of the i th atom, r_{ij} represents the interatomic distance, Z_i is the atomic charge, ϵ_0 is the vacuum permittivity, $\Phi(r_{ij}/a)$ is the screening function in the ZBL potential with a screening length a , and $S(r_{ij})$ is the switching function. The screening function $\Phi(x)$ is expressed as $0.18175e^{-3.19980x} + 0.50986e^{-0.94229x} + 0.28022e^{-0.4029x} + 0.02817e^{-0.20162x}$ and the screening length a is defined as $0.4685/(Z_i^{0.23} + Z_j^{0.23})$ in Å.³⁴ The switching function $S(x)$ in eq 3 ensures a smooth decay of the ZBL potential to zero at finite distances, using a fourth-order polynomial (details are available in the LAMMPS documentation³⁵). This function introduces two cutoff distances: the inner cutoff (r_c^{in}) and the outer cutoff (r_c^{out}). At $r_{ij} = r_c^{\text{out}}$, $S(x)$ cancels out the screened Coulomb potential such that $E_{ij}^{\text{ZBL}} = 0$. For $r_{ij} < r_c^{\text{in}}$, $S(x)$ becomes a constant, contributing zero to the force. The parametrization of $S(x)$ ensures that the ZBL potential is smooth up to the second derivative at r_c^{in} and r_c^{out} . We tune the cutoff radii for each pair of atomic species (Si, N, H, and F) as follows: the equilibrium distance (d_{eq}) is first obtained for each dimer in vacuum

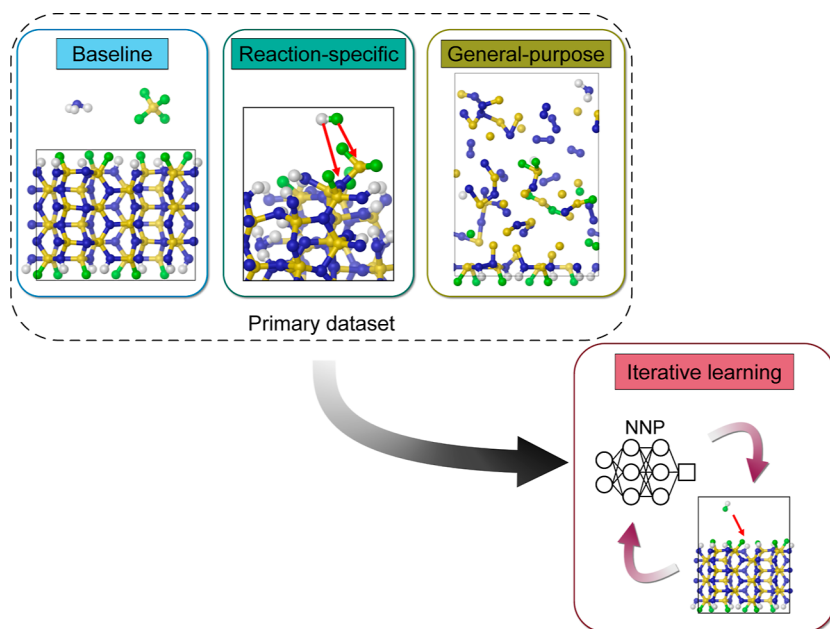


Figure 1. Schematic of the data set construction for NNP training. Initial baseline structures cover diverse equilibrium configurations. Reaction-specific sets are captured via guided MD, while general-purpose sets are derived from high-temperature MD simulations. Iterative refinements of the NNP are conducted using trajectories of NNP-based MD simulations.

through DFT calculations. The cutoff distances r_c^{in} and r_c^{out} are then set to $0.9 \times d_{\text{eq}}$ and $0.85 \times d_{\text{eq}}$, respectively. (See Table S1 for actual values.)

To train hybrid-style NNP in eq 1, we utilize the residual energy, which is obtained by subtracting the ZBL potential from the DFT energy. The energy loss function (Γ_E) in batch training is given as follows

$$\Gamma_E = \frac{1}{M} \sum_{i=1}^M \left[\frac{(E_{\text{DFT}}^i - E_{\text{ZBL}}^i) - E_{\text{NNP}}^i}{N_i} \right]^2 \quad (4)$$

where M is the number of structures in a batch, N_i is the number of atoms in the i th structure, and E_{DFT}^i (E_{ZBL}^i , E_{NNP}^i) is the DFT (ZBL, NNP) energy of the i th structure. The loss functions for force and stress are defined similarly for each component.

We utilize SIMPLE-NN package to train the Behler–Parrinello-type NNP, which employs atom-centered symmetry functions (ACSFs) as the input features.³⁶ The numbers of radial and angular features for both Si and N are 24 and 96, respectively, while corresponding numbers for both H and F are 16 and 60, respectively. The cutoff radius for ACSF is 4.5 Å except for 6 Å for the radial features between Si and N. Further details of ACSFs are compiled in Table S2. As a preprocessing, the descriptors are first scaled using the maximum and minimum values for each ACSF and then decorrelated using principal component analysis with whitening. The fully connected neural network has two hidden layers with 60 nodes each and uses a hyperbolic tangent activation function. The model is L2-regularized with a coefficient of 10^{-6} . We split the structures of the data set into 90% for the training set and 10% for the test set. The training is carried out in two steps, each with 100 epochs, but differing in learning rate and minibatch size. The first step employs a learning rate of 10^{-4} and a batch size of 4, while the subsequent step uses a learning rate of 10^{-5} and a batch size of 24. For optimization, we utilize the Adam optimizer.³⁷

2.2. DFT Calculations. In constructing training data, we carry out DFT computations by using the Vienna Ab initio Simulation Package (VASP).³⁸ The Perdew–Burke–Ernzerhof (PBE) functional is used to approximate the exchange–correlation energy among electrons.³⁹ To optimize computational efficiency, we initially perform ab initio molecular dynamics (AIMD) simulations with a low-fidelity setup employing soft-core pseudopotentials for N and F and unpolarized

spin configurations. Subsequently, these trajectories are subsampled and recalculated with a high-fidelity setting, which incorporates standard pseudopotentials, a higher cutoff energy of 500 eV, and spin-polarized functional. In the supercell calculations, only the Γ point is sampled for the Brillouin zone integration and the temperature is controlled by the Nosé–Hoover thermostat.⁴⁰ During the AIMD, the hydrogen mass is tripled to enable a larger time step of 1 fs, while ensuring the stability of the simulations at high temperatures. In cases where hydrogen is absent in the AIMD simulations, the time step is set to 2 fs.

2.3. MD Simulation for Etching. We conduct MD simulations of etching processes using an NNP within the LAMMPS software.⁴¹ The etching simulation involves the repeated impact of HF molecules on a substrate in the xy plane. Each impact cycle introduces a randomly oriented HF molecule with an initial velocity (v_{ini}), starting from a position that is random along the lateral directions and at least 6 Å above the substrate surface. The azimuth angle of v_{ini} is chosen randomly, while the incident energy (E_{in}) and the angle of incidence (θ_{in})—defined as the angle between v_{ini} and the negative of the surface normal—are specifically selected for each simulation.

After each introduction of an HF molecule, the simulation proceeds in the NVE ensemble for 2 ps, followed by an NVT simulation at 300 K using the Nosé–Hoover thermostat. Once the temperature stabilizes at 300 K, the next HF molecule is introduced. To prevent the substrate slab from drifting after impacts, the bottom layer with a thickness of 2 Å is fixed.

During the bombardment by ions, high velocities or accelerations of atoms can introduce errors into the velocity Verlet algorithm. To mitigate these errors and save computational resources, we adopt a variable time step (Δt).⁴² The value of Δt for each MD step is determined under three conditions: the atom displacement does not exceed 0.1 Å, the kinetic energy transfer is limited to 1 eV, and Δt does not exceed 0.5 fs.

The van der Waals (vdW) interactions are crucial for interactions between the surface and adsorbed gas molecules, significantly affecting desorption rates of gas molecules that are weakly bound to the surface. Neglecting vdW interactions could lead to an inaccurate representation of molecular behaviors, such as exaggerated desorption rates. Since the reference DFT data sets are obtained within PBE, which poorly describes vdW interactions, the vdW contribution is separately added to E_{pot} . Thus

$$E_{\text{pot}} = E_{\text{ZBL}} + E_{\text{NNP}} + E_{\text{vdW}} \quad (5)$$

where E_{vdW} is the vdW energy according to the DFT-D2 formalism.⁴³ The cutoff distance for the vdW interaction is set to 15 Å.

3. RESULTS AND DISCUSSION

3.1. Training Neural Network Potential. The objective of this study is to investigate the etching process of a-Si₃N₄ by HF molecules using MD simulations. To develop an NNP suitable for the simulation, constructing comprehensive training sets that accurately reflect the chemical reactions in the HF etching process is crucial. However, identifying all possible chemical reactions exhaustively is computationally infeasible. To address this challenge, we propose a strategy that involves constructing diverse training sets along with an iterative training for further refinements. (See Figure 1.). An overview of the training sets is presented in Table 1. In the below, we provide details in the training sets, iterative training, and further verification of NNP.

Table 1. Overview of the Training Set configurations

data type	description	training points
baseline	molecules (NH _x F _y , SiH _x F _y)	4478
	NVT MD (1000 K)	
	crystalline bulk of α-Si ₃ N ₄ , β-Si ₃ N ₄	26 642
	NVT MD (300 K), strained cells	
	crystalline β-Si ₃ N ₄ slab (0001)	21 600
	NVT MD (300, 2000 K)	
	Si/N interstitial, vacancy in a β-Si ₃ N ₄ slab	141 300
	NVE MD	
	channeling in β-Si ₃ N ₄ bulk, slab	54 750
	NVT MD (2000 K) with 10 eV HF	
	liquid, amorphous Si ₃ N ₄ bulk	114 912
	melt-quench-annealing trajectories	
	amorphous slab	70 000
	NVT MD (500 K, 3000 K)	
	liquid HF	25 000
reaction-specific	NVT MD (300 K, 500 K)	
	guided MD: HF guided to the β-Si ₃ N ₄ slab	558 666
general-purpose	constrained MD at 2000 K	
	high temperature-MD of a β-Si ₃ N ₄ slab	492 000
	NVT MD (4500 K, 10,000 K)	
iterative learning	subsampled NNP MD trajectories	308 208
	$E_{\text{error}} \geq 15$ meV/atom	

3.1.1. Training Data Set. The training sets are divided into three categories: baseline, reaction-specific, and general-purpose. We detail each category in the below.

3.1.1.1. Baseline Structures. The baseline structures include a diverse range of elementary structures, such as radicals, molecules, and both crystalline and amorphous forms of bulk and slab Si₃N₄. To sample the local potential energy surface of these structures, AIMD simulations at various temperatures are carried out. In particular, AIMD at elevated temperatures is beneficial for ensuring the stability of the NNP.²⁸

The molecular subset of our training set encompasses single atoms and every possible diatomic combination of Si, N, H, and F. It also incorporates polyatomic molecules such as NH_xF_y ($0 \leq x, y \leq 3$; $0 < x + y \leq 3$) and SiH_xF_y ($0 \leq x, y \leq 4$; $0 < x + y \leq 4$). This aims to cover a wide range of potential byproducts, spanning from stable molecules to metastable

radicals. The polyatomic molecules (NH_xF_y and SiH_xF_y) are sampled during MD at 1000 K for 2 ps while the diatomic molecules are sampled at discrete bond lengths.

For crystalline structures, α and β phases of Si₃N₄ are considered with various strains and NVT–MD simulations at 300 K for 2 ps. To obtain surface-related training data, we construct an orthorhombic slab structure of β-Si₃N₄ with dimensions of 7.6 × 13.3 × 18.7 Å³, exposing the (0001) plane with Si–F and N–H terminations. The local vibrations are sampled at temperatures of 300 and 2000 K, for durations of 2 ps each.

To consider primitive defects generated during etching, we introduce both vacancies and interstitials of Si and N into the slab separately. Vacancies are created in the subsurface layer, taking crystallographic equivalence into account, while interstitials are positioned at the centers of both large and small channels. The training data are obtained from MD simulations conducted at 2000 K, with durations ranging from 4 to 8 ps, until the local structures reach a steady state following reconstruction. Additionally, the incoming HF molecules can traverse the channels along the *c*-axis in β-Si₃N₄. To account for this, we sample 5 ps MD trajectories in which an HF molecule passes through both slab and bulk channels with microcanonical ensemble. The HF molecule is initially placed at the center of the channel cross section with a kinetic energy of 10 eV.

The bulk liquid and amorphous Si₃N₄ are sampled from melt-quench trajectories. The melt-quench simulation is carried out as follows: 48 Si and 64 N atoms are randomly distributed in a cubic cell with the length of 1.06 nm. The resulting density of 3.11 g/cm³ corresponds to the experimental density of a-Si₃N₄.⁴⁴ This cell is first premelted at 4000 K (this stage is not included in the training set) and then maintained in a liquid state at 3000 K for 20 ps. Subsequently, it is quenched to 300 K at a rate of −100 K/ps, and the resulting amorphous structure is annealed at 500 K for 15 ps. We then randomly cleave the bulk amorphous Si₃N₄ to form a slab structure and the slab is melted by increasing the temperature from 500 to 3000 K at a rate of 250 K/ps. Various surface configurations are sampled at 500 K for 5 ps and 3000 K for 20 ps (the MD trajectory, where temperature increases to 3000 K, is not sampled).

During the simulation, unreacted HF molecules can accumulate and aggregate on the substrate, forming hydrogen bonds among HF molecules as in a liquid phase. To account for this, the HF liquid is simulated for 10 ps with NVT ensemble at 300 and 500 K and a density of 0.971 g/cm³ (density at 292.9 K).⁴⁵

3.1.1.2. Reaction-Specific Structures. In addition to the aforementioned bulk and slab configurations, the training set should also encompass potential chemical reactions. These are not easily sampled during conventional MD simulations as chemical reactions are rare events with significant activation barriers. Here we utilize a guided MD approach to effectively sample reaction pathways. This technique can accelerate and sample reactions within the MD time scale by employing constrained MD and gradually reducing the constraints until the desired configuration is attained.^{46,47} We take advantage of the observation that H tends to bond with N, while F prefers to bond with Si. This bonding preference is supported by the experimental analysis on surface states following etching with HF solution.⁴⁸ Reflecting this, we conduct guided MD

simulations at 2000 K by applying a constraint to the collective variable S defined as follows

$$S = R_{\text{Si-F}} + R_{\text{N-H}} \quad (6)$$

where $R_{\text{Si-F}}$ ($R_{\text{N-H}}$) is the distance between Si (N) and the targeted F (H) atoms. S is gradually reduced at a rate of 0.02 Å/fs until it reaches 2.8 Å, which corresponds to 1.05 times the sum of bond lengths in SiF_4 and NH_3 (1.63 and 1.03 Å, respectively).

One example of guided MD is depicted in Figure 2. Initially, a HF molecule is placed 4 Å above the target Si–N bond, and

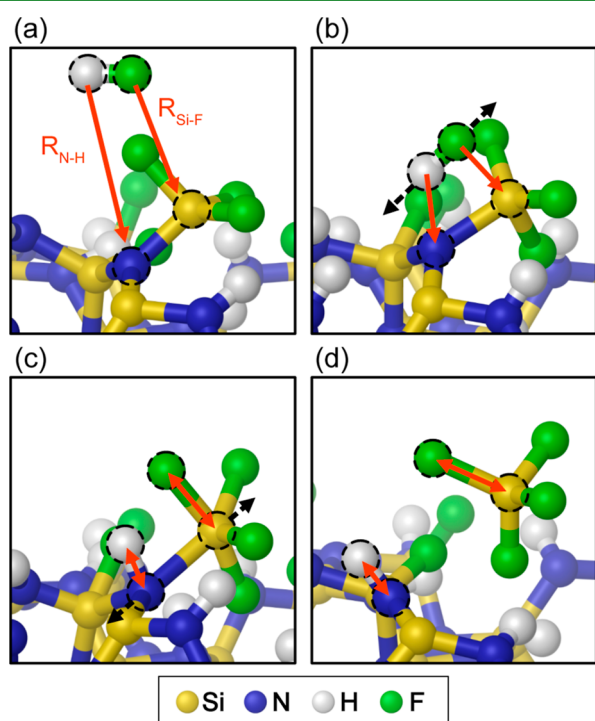


Figure 2. Schematic diagram of guided MD: (a) an HF molecule approaching the target Si–N bond; (b) breaking of the HF bond; (c) SiF_4 in a transition state before desorption; (d) SiF_4 departing from the surface.

its orientation is set randomly (Figure 2a). The collective variable S is gradually reduced, and once S reaches 2.8 Å (Figure 2b), the MD simulation is continued for 5 ps, with S maintained constant (Figure 2c,d). It is observed that a SiF_4 gas molecule forms as a result of dissociative reaction of HF. Given that the reaction energy and barrier are primarily influenced by the first neighbors around the target Si–N bond, we sample a total of 40 configurations with diverse local environments. About 10% of the target Si–N bonds are not broken after the guided MD simulation. (These are still included in training sets.)

3.1.1.3. General-Purpose Data Set. During collision cascades from high-energy impacts, a wide range of unstable configurations with low coordination numbers can be generated, which may not be sufficiently sampled in the baseline or reaction-specific data set obtained under relatively mild conditions. In addition, the high-temperature AIMD can help to increase accuracy in unexpected structures.²⁸ Therefore, we construct a general-purpose data set that samples configurations from high-temperature MD simulations of Si_3N_4 slab, which results in active bond breaking and formations. We

consider three types of slab structures: the first one is the cleaved crystalline slab structure, the same structure as in baseline structures. The other two are created by removing either two Si atoms or two N atoms from the top surface and passivating the dangling bonds around N and Si with H and F atoms, respectively. The variation in composition is implemented to sample different disordered structures. The bottom two layers of the slab are fixed, including the terminating atoms, so as to partially maintain the crystal structure at high temperatures. This approach would allow the sampling of approximate interfaces between highly disordered structures and the stoichiometric bulk during collisions. The sampling temperatures are 4500 and 10,000 K, using the NVT ensemble for 10 ps. Figure S1 illustrates the structures from high-temperature MD at 10,000 K which contain disordered surfaces of Si_3N_4 and diverse molecules. We note that the general-purpose training set is crucial because eliminating this set results in unstable MD simulations during etching process (results not shown).

With the above-mentioned training sets (3.1.1.1–3), we train a primitive NNP. The root-mean-square error (RMSE) for the test set is 18 meV/atom for energy, 0.64 eV/Å for force, and 8.6 kBar for stress. (The parity plots are provided in Figure S2a–c). As a basic test, we compare structural properties of the a- Si_3N_4 obtained by the NNP with those obtained using DFT in Figure S2d,e. It is observed that the NNP well reproduces the radial and angular distributions.

3.1.2. Iterative Training. To evaluate and refine the primitive NNP, we carry out small-scale etching simulations using the primitive NNP on crystalline and amorphous Si_3N_4 surface models and obtain the DFT data for subsampled configurations, which are used for retraining the NNP iteratively. For the crystalline slab, we employ the same structure as in the baseline. To generate an amorphous slab, we utilize a melt-quench-anneal protocol using the NNP. In detail, 42 Si and 56 N atoms are first randomly distributed in a cubic cell with the length of 1 nm. This is followed by premelting at 5000 K for 20 ps and melting at 2800 K for 100 ps. The system is then quenched to 300 K at a rate of -10 K/ps, and annealed at 500 K for 20 ps. The amorphous structure is randomly cleaved, and the obtained slab is annealed at 300 K for 5 ps.

In the etching simulations, we follow the procedures outlined in Section 2.3 except that the vdW interaction is not included for consistency of the functional with the primary data set. Each HF molecule is assigned a random E_{in} and θ_{in} within the ranges of 10–50 eV and 0–75°, respectively. After the last HF molecule impacts, DFT calculations are performed on trajectories selected within the initial 2000 steps after each incidence (~ 1 ps), which are densely sampled at intervals of 40 steps. Subsequent trajectories are sampled at a less frequent interval of 200 steps. Structures where the energy difference exceeds 15 meV/atom between NNP and DFT are incorporated into the training sets, and the NNP is retrained. The update of NNP is iterated through three cycles each for the crystalline and amorphous slabs, with each cycle beginning anew from a clean slab. The first iteration includes 25 incidences of HF molecules, while each subsequent iteration involves up to 50 HF molecule incidences.

Figure 3a illustrates the NNP and DFT energies along the MD trajectory for the first cycle of iterative training (before any refinements), which consists of 1533 subsampled configurations. It is seen that the NNP energy closely matches the DFT energy throughout the trajectory. The RMSEs of

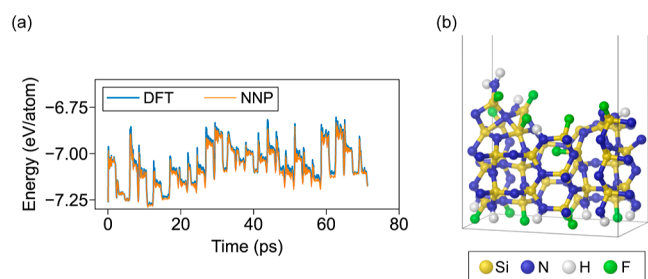


Figure 3. (a) Comparison of the DFT and NNP energies along the NNP MD trajectory for the crystalline slab after the primitive training and (b) the etched surface after the NNP MD simulation.

energy and force for these configurations are 16.7 meV/atom and 0.65 eV/Å, respectively, which are comparable to those from the test set of the primary data set (18 meV/atom and 0.64 eV/Å). The pronounced peaks in Figure 3a indicate moments of close impact, with the sustained high energy levels related to heating of the slab up to ~ 2500 K. The subsequent reductions in energy correspond to the cooling by the thermostat. After 25 HF incidences, 2 Si and 3 N atoms are etched away and the final configuration is shown in Figure 3b.

In subsequent iterations of both the crystalline and amorphous slabs, the trends in NNP and DFT energy are similar to those shown in Figure 3. The RMSE values for energy and force across the trajectories are within the ranges of 10–17 meV/atom and 0.59–0.72 eV/Å, respectively. These values do not seemingly represent significant improvement in the accuracy of NNP, but the iterative training helps to increase the accuracy in reaction energies, which will be discussed in Section 3.1.3.

3.1.3. Test on Reaction Energies. In the above, the accuracy of the NNP was examined by comparing it with DFT results for the sampled snapshots during small-scale etching MD. However, comparison of the total energy or atomic forces in large supercells can be obscured by thermal vibrations that exist throughout the whole system. To more directly compare the energy landscape for specific chemical reactions during the etching process, we first identify stable configurations during the etching MD with the crystalline slabs in the previous subsection. These are defined as snapshots where the topology of the bond network is well maintained over 100 consecutive steps. Here, two atoms are assumed to be bonded if the interatomic distance is less than 1.3 times the sum of their covalent radii. We identify 1 to 5 distinct stable states for each incidence and relax them using NNP and DFT to remove thermal fluctuations. The vdW interactions are omitted here to validate the accuracy of the NNP against PBE as the reference data are based on PBE.

We define the reaction energy (E_{rxn}) as the energy difference between two consecutive and distinct stable states. The energies for gaseous reactants and products are also considered. In total, 212 reactions are identified and their E_{rxn} values are compared between DFT ($E_{\text{rxn}}^{\text{DFT}}$) and NNP ($E_{\text{rxn}}^{\text{NNP}}$) in Figure 4. A good correlation between $E_{\text{rxn}}^{\text{DFT}}$ and $E_{\text{rxn}}^{\text{NNP}}$ is found with the coefficient of determination (R^2) of 0.88. These reactions encompass various processes, including bond breaking and rearranging Si–N bonds in the substrate, the bond formations between incoming molecules and the substrate, and desorption of byproducts. The results suggest that the refined NNP accurately describes a wide range of chemical reactions. We note that the primitive NNP without

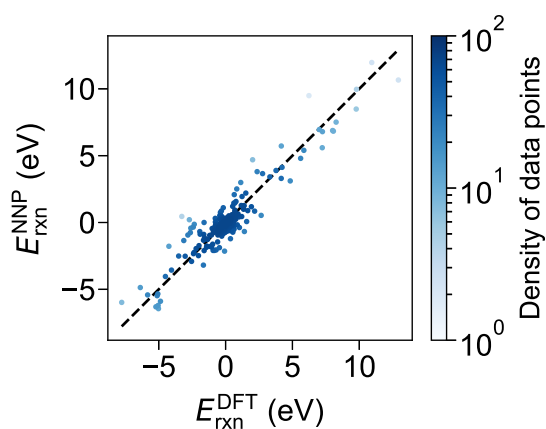


Figure 4. Comparison of reaction energies during etching MD calculated using DFT ($E_{\text{rxn}}^{\text{DFT}}$) and NNP ($E_{\text{rxn}}^{\text{NNP}}$). Each point refers to individual reactions. The density of data points on the color bar indicates the local density of reaction events in the vicinity of each point, accounting for the superposition of many data points.

iterative training exhibits a R^2 value of 0.81, suggesting significant improvement in the reaction energy after the iterative training.

3.2. Etching Simulations. We apply the NNP trained through the previous subsection to simulating etching process in larger a-Si₃N₄ models with a surface area of 2×2 nm², which are generated by following the melt-quench-annealing protocol in Section 3.1.2. The pristine a-Si₃N₄ slabs are used based on the assumption that the initial termination does not affect the etching behavior in steady state. To explicitly show this, we also test with amorphous slabs with hydrogen termination, and confirm similar results with those from the pristine model. (See Table S3 and Figure S3 for details.)

Total 800 HF molecules are incident sequentially (see Section 2.3), which is equivalent to an ion dose of 200 HF/nm², over 3–6 ns. We note that the average molecular flux ($\sim 10^{10}$ HF/nm²·s) is roughly 8 orders of magnitude higher than the typical experimental condition in plasma etching.⁴⁹ Nevertheless, the interaction time of ions is known to be on the order of picoseconds,⁷ so important dynamics can be simulated within the MD time scale (~ 5 ps/incidence in this study). For example, a similar MD simulation produced etching yields of SiO₂ and SiN by CF₃⁺ ions that agree well with experimental results.⁵⁰ However, long-term reactions such as densification or annealing may not be fully simulated with the current MD protocol, which might result in differences in etching yield and chemical composition at surfaces. Such effects are of interest for future study.

The simulations encompass a range of E_{in} (from 10 to 50 eV) at normal incidence, as well as variations in θ_{in} (from 0 to 75°) at a fixed energy of 50 eV. For each incidence, θ_{in} is varied following a Gaussian distribution with a standard deviation of 5° to account for the typical etching condition.⁵¹ For statistical average, three simulations are carried out with distinct amorphous cells for each energy–angle combination.

We assume that stable byproducts such as SiF₂, SiF₄, N₂, NH₃, HF, H₂, and F₂, easily desorb from the surface owing to the low-pressure conditions of plasma etching. These products that are not bonded to the substrate are identified by analyzing the topology of bond networks and removed after each cycle of molecule injection. (Two atoms are assumed to be bonded if their interatomic distance is less than 1.15 times the sum of

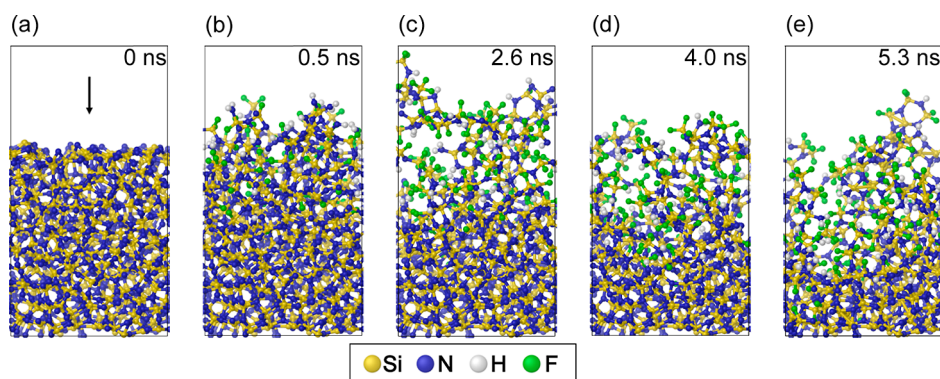


Figure 5. (a–e) Snapshots of the etching MD progress at $t = 0, 0.5, 2.6, 4.0, 5.3$ ns. The HF molecules are incident vertically with the kinetic energy of 50 eV.

their covalent radii.) We confirmed that this procedure has minimal influence on the forces around the products by checking the average magnitude of the force difference after product removal, which is an order of magnitude smaller than the average magnitude of the original forces.

3.2.1. Etching Process at Normal Incidence. We first examine the etching results when HF molecules, with E_{in} of 50 eV, strike the surface at normal incidence ($\theta_{\text{in}} = 0^\circ$). During the etching process, the incident HF molecules undergo collision cascades, leading to three potential outcomes: reflection from the surface, formation of etching products, or implantation within the slab substituting Si–N bonds with Si–F and N–H bonds. Figure 5 illustrates the sequence of surface evolution during etching. Initially, HF molecules break Si–N bonds, modifying the surface layer rather than etching. This modification reduces the atomic density in the surface layers (Figure 5b,c). Once the modified layer reaches a certain thickness, a dynamic equilibrium is established, and the height of the slab begins to decrease (Figure 5d,e).

The variation in the number of each species with respect to the HF dose is depicted in Figure 6a. Notably, the counts of H

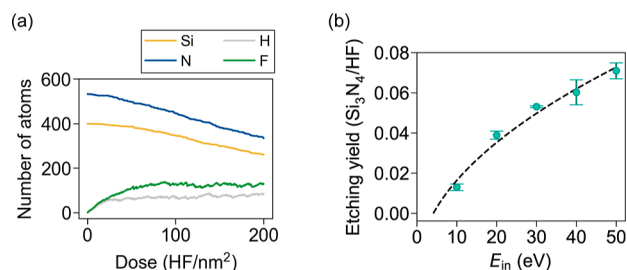


Figure 6. (a) The number of Si, N, H, and F atoms in the simulation cell during the etching process at $E_{\text{in}} = 50$ eV and $\theta_{\text{in}} = 0^\circ$. (b) Etching yield as a function of incident energy. The black dashed line represents the fitted function of the universal etching behavior.⁵² The error bars indicate the standard deviations among three independent runs.

and F atoms rise until approximately 100 HF/nm² (or $t = 2$ ns), and then stabilize. This suggests that, initially, H and F atoms are more likely to be implanted in the slab and contribute to the formation of the modified layer rather than being reflected or contributing to etching products. Conversely, the etching yield for Si and N increases until the dose reaches 100 HF/nm². Particularly, the etching yield of N compared to Si is greater than the stoichiometric ratio of 1.33,

before a steady state is established at 100 HF/nm², indicating an early stage depletion of nitrogen at the surface. This will be discussed in more detail below. Both Si and N etching yields appear to stabilize after 100 HF/nm², coinciding with the constant H and F quantities. We calculate the steady-state etching yields of Si and N from the slopes between 100 and 200 HF/nm². The average etching yields of Si and N, derived from three distinct amorphous slabs, are 0.216 and 0.281 atoms/HF, respectively. The ratio of these yields (1.3) is close to the stoichiometric ratio of 1.33 for Si₃N₄. Note that experimental etching yields by CF₃⁺ and Ar⁺ ions at 50 eV are extrapolated to be 0.26 and 0.0219 Si atoms/ion, respectively.⁵³

The depletion of nitrogen at the surface is consistently observed across a range of incident energies. (See Figure S4). The nitrogen depletion was also noted in several experiments on SiN etching.^{54,55} Based on the MD trajectory, we attribute this preferential sputtering of N to the facile formation of N₂. This occurs when two N atoms, freed from their bonds due to ion collision, come into proximity and form a stable N₂ molecule. This newly formed N₂ may weakly chemisorb or physisorb on the surface, desorbing with minimal energy gain from subsequent ion impacts. Indeed, N₂ is a common byproduct in various plasma studies.^{54–58} We anticipate that plasma etching will reduce the nitrogen content of surfaces, potentially leading to an excess of Si atoms and a degradation in electrical properties.^{59,60}

In Figure 6b, total etching yields (Y) in the unit of Si₃N₄/HF are plotted with respect to E_{in} . We fit the data to the well-known formula for Y ⁵²

$$Y(E_{\text{in}}) = A(\sqrt{E_{\text{in}}} - \sqrt{E_{\text{th}}}) \quad (7)$$

where E_{th} is the threshold energy and A is a fitting constant. The fitted values for A and E_{th} are 0.0144 and 4.1 eV, respectively, and the fitted function is depicted as a dashed line in Figure 6b. This indicates that the kinetic energy of HF should be at least 4.1 eV to initiate surface modification and reaction to form products in the RIE of Si₃N₄.

To further understand the etching process, we analyze the desorbed byproducts. Figure 7 shows the number of desorbed species containing Si or N, counted within a dose range of 100 to 200 HF/nm². The primary etching products are SiF₂, SiF₄, NH₃, N₂, and clusters of Si_xN_yH_nF_m (hereafter referred to as Si_xN_y). The predominance of SiF₂ and SiF₄ in Figure 7 is similar to that observed in the main products of experiments on SiN etching using XeF₂ or fluorocarbon-based plas-

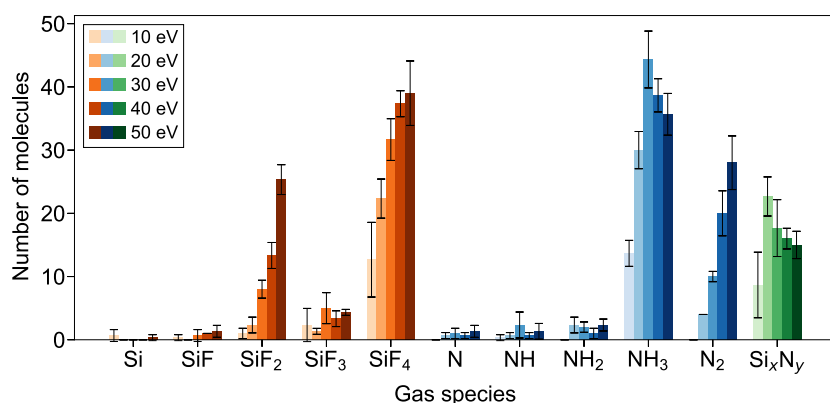


Figure 7. Number distribution of etching products with regard to E_{in} . The error bars indicate the standard deviations among three independent runs.

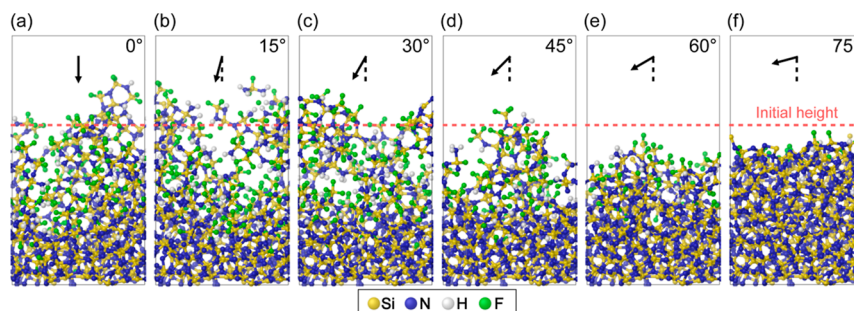


Figure 8. (a–f) The etched surfaces of a- Si_3N_4 with regard to the angle of incidence (θ_{in}) of HF: $\theta_{in} = 0, 15, 30, 45, 60$ and 75° , respectively. The kinetic energy of the HF molecule is 50 eV. The initial height of the surface before etching is noted as a dashed horizontal line.

mas.^{58,61,62} SiF_2 , despite being a radical, is a prevalent product due to its unusually long half-life.⁶³ Additionally, a small amount of fluorosilane (SiH_nF_m), in particular SiHF_3 emerge as byproducts. They are implicitly included in the corresponding SiF_x category for simplicity.

In Figure 7, we observe a significant presence of Si_xN_y clusters. Given their larger molecular weight compared to other products, they are more prone to redeposition. This tendency toward redeposition could potentially reduce the etching yield in experimental setups or influence sidewall deformations. It is also observed that a higher E_{in} increases the relative portion of SiF_2 and N_2 among the etching products. This trend is consistent with experimental observations indicating that unsaturated species, such as SiF_2 , become more abundant with increased ion energy during the etching of silicon and its oxide.⁶⁴ Regarding N_2 , higher E_{in} results in more intense collision cascades upon the surface, enhancing the chance for N atoms to form N_2 . We find that the desorption of SiF_3 does not increase with the E_{in} due to its high reactivity. That is to say, most SiF_3 are redeposited onto the substrate before desorption, or desorbed as SiF_2 or SiF_4 after reacting with the substrate.

To investigate the size effect of the periodic boundary condition on the etching yield, we conduct etching simulations using two larger models with areas of $2.8 \times 2.8 \text{ nm}^2$ and $3.5 \times 3.5 \text{ nm}^2$. These models are prepared using the same method as the $2 \times 2 \text{ nm}^2$ slab. The slabs are then exposed to HF bombardments at 50 eV and normal incidence. The resulting atomic density profiles of the substrates are similar as shown in Figure S5. The etching yields, averaged over three independent runs, are 0.0735 and 0.0663 $\text{Si}_3\text{N}_4/\text{HF}$ for the $2.8 \times 2.8 \text{ nm}^2$ and $3.5 \times 3.5 \text{ nm}^2$ models, respectively. These yields are

similar to the 0.0708 $\text{Si}_3\text{N}_4/\text{HF}$ observed for the $2 \times 2 \text{ nm}^2$ slab, with all results within a 5% variance. This consistency indicates that the $2 \times 2 \text{ nm}^2$ slab model is sufficiently large to avoid artifacts from the periodic boundary condition.

3.2.2. Angular Dependence of Etching Process. Next, we present the results of a- Si_3N_4 etching under various angles of incidence, ranging from vertical (0°) to nearly horizontal orientations (75°), while E_{in} is fixed at 50 eV. Figure 8 depicts the etched structures after a dose of $200 \text{ HF}/\text{nm}^2$ with selected θ_{in} . Notably, as θ_{in} increases, the modified layer becomes progressively thinner. At 75° , the modified layer is virtually absent, with the uppermost surface atoms being passivated by H and F atoms (see Figure 8f). To explain this, we plot the reflection coefficients in Figure 9a, which are defined as the ratio of HF molecules reflected without implantation or contributing to etching products. It is seen that the reflection

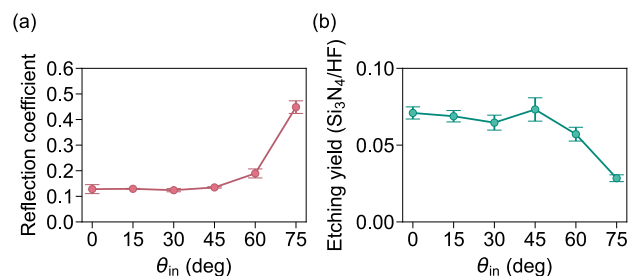


Figure 9. (a) The reflection coefficient, and (b) the total etching yield per incident HF molecule, both as a function of the incident angle. The error bars indicate the standard deviations among three independent runs.

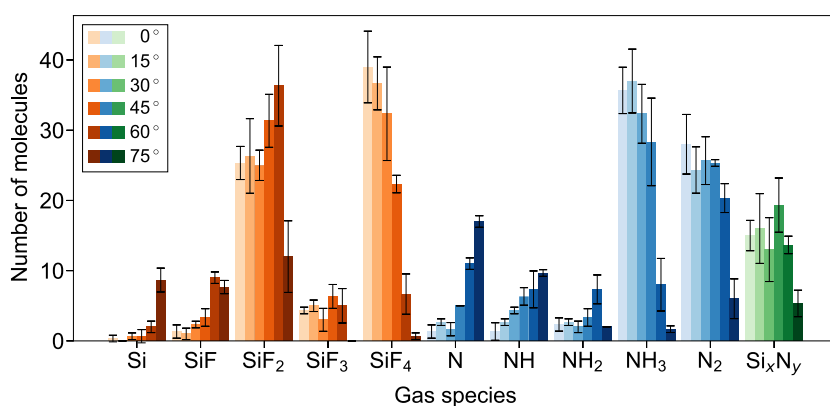


Figure 10. Number distribution of etching products with regard to θ_{in} . The error bars indicate the standard deviations among three independent runs.

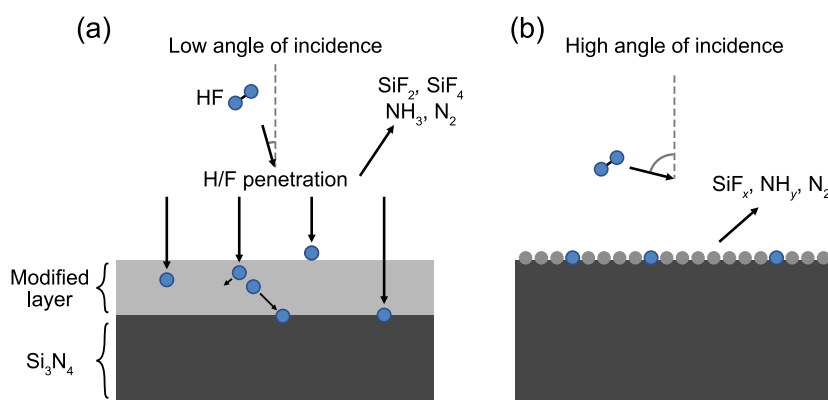


Figure 11. Schematic of atomistic mechanism in Si_3N_4 etching (a) at low angle of incidence, and (b) high angle of incidence.

coefficient increases with θ_{in} . That is to say, incident HF molecules are more likely to be reflected from the surface at high θ_{in} , rather than penetrating into the substrate, which is necessary for developing the modified layer. Consequently, the surface modification is suppressed for high θ_{in} values. Furthermore, the vertical component of the kinetic energy of the incoming HF molecules, which relates to the penetration depth, decreases with θ_{in} . This results in shallower penetration of the molecules, further limiting the growth of the modified layer. To note, it was also observed in experiments that ions show higher probabilities of reflection at increased θ_{in} values.^{65,66}

Figure 9b shows the etching yield as a function of θ_{in} . The etching yield decreases rapidly at high θ_{in} values because the reflection coefficient increases and the surface is less modified. It is notable that the etching yield exhibits a peak at 45° . This is a result of the interplay of physical sputtering and chemical reaction processes, known as chemical sputtering.^{67,68} Chemical sputtering is further discussed in the analysis of desorbed products below.

Figure 10 shows the number of desorbed species at different incident angles. At lower angles, ranging from 0 to 45° , the primary products are saturated molecules like SiF_4 , NH_3 , and N_2 . This suggests a chemical sputtering process where reactive atoms that have been implanted help in forming the product molecules. Conversely, at higher angles, the product species shifts significantly toward under-coordinated molecules or radicals, such as Si, SiF, N, and NH. This is due to a reduced presence of H and F atoms on the surface at higher angles, which limits the formation of saturated molecules. As a result,

the role of chemical reactions in the overall etching process lessens as surface modifications decrease. A notable finding is the reduced population of SiF_2 at 75° . Although SiF_2 usually forms easily and desorbs energetically with sufficient surface alteration, it is suppressed because F atoms are depleted at the surface.

The slight increase in etching yield at 45° in Figure 9b and the creation of unsaturated radicals in Figure 10 are both results of the combined effects of physical sputtering and chemical reactions. The incoming HF molecules can most efficiently knock out etching byproducts when they strike the surface at oblique angles (physical sputtering). Conversely, reactive atoms are more effectively implanted at shallow angles of incidence, where they interact with the surface to weaken the surface bonds. This exemplifies the dual nature of etching yield observed in chemical sputtering.^{67,68} It can be anticipated that higher kinetic energy will yield a stronger peak at the oblique angle due to physical sputtering, as observed in the previous experiment.⁶⁹

3.2.3. Etching Mechanism. Based on the observations on surface species and etching products in the above, we arrive at a schematic etching mechanism as illustrated in Figure 11. At low incident angles, HF molecules can penetrate the Si_3N_4 substrate, leading to the formation of modified surface layers. Within these layers, Si–N bonds are progressively replaced by Si–F and N–H bonds. The abundant H and F atoms facilitate the formation of fully coordinated molecules such as SiF_4 and NH_3 . Concurrently, a deficiency in N arises due to preferential sputtering. In contrast, at higher incident angles, the implantation of H and F into the surface is reduced. Given

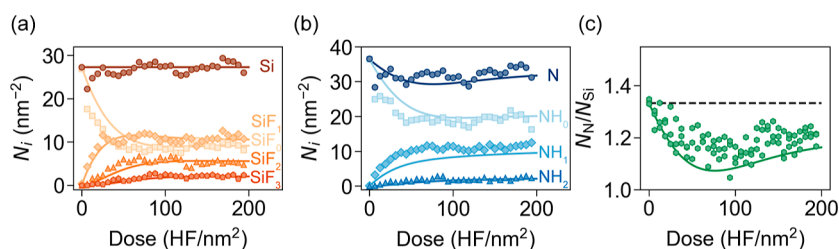
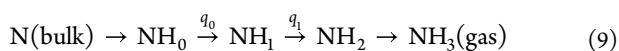
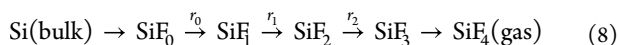


Figure 12. Evolution of populations of surface species (N_i) in the modified layer as a function of HF dose. The solid lines are the results of 1D continuum model while symbols are directly obtained from MD simulations with E_{in} of 50 eV and θ_{in} of 0° . Panel (a) depicts the population change of silicon fluoride species, namely SiF_0 , SiF_1 , SiF_2 , and SiF_3 . Panel (b) illustrates the changes in ammonia-derived species, specifically NH_0 , NH_1 , and NH_2 . Panel (c) presents the ratio of N to Si, with the dashed line indicating the bulk stoichiometry of 1.33.

the challenge for surface Si and N atoms to scavenge H and F atoms under these conditions, they are more likely to desorb as unsaturated radicals, which characterizes physical sputtering for high angles of incidence.

3.3. Bridging to Mesoscale Model. In the previous section, the atomistic MD simulation revealed the finest details of the etching process. While this provides the most accurate description of the process, the size and time scales are far smaller than those required for realistic simulations. To address this, it is necessary to bridge the gap between atomistic simulation and mesoscale models through a coarse-graining procedure. Here, we demonstrate such scale-bridging utilizing our simulation results for the specific case where HF is incident vertically with the kinetic energy of 50 eV.

Based on the MD simulations, we make the following assumptions about the surface states during the etching process: First, Si and N are always four- and three-fold coordinated, respectively, owing to the strong covalent nature of the Si–N, Si–F, and N–H bonds. Second, the incidence of HF can increase the number of Si–F and N–H bonds stepwise or etch surface species with certain probabilities that are constant under the given incidence condition. Thus, the chemical reaction proceeds as follows



where SiF_n ($n = 0, 1, 2, 3$) and NH_m ($m = 0, 1, 2$) represent Si and N atoms within the modified layer, which are bonded to n F and m H atoms, respectively. (The remaining bonds are all Si–N bonds.) In eqs 8 and 9, r_n and q_m denote the reaction probabilities from SiF_n to SiF_{n+1} and from NH_m to NH_{m+1} per unit dose of HF, respectively. Additionally, we note desorption probabilities or partial etching yields of each species per unit HF dose in terms of Γ_{SiF_n} ($n = 0, 1, 2, 3$) and Γ_{NH_m} ($m = 0, 1, 2$). For simplicity, the desorption probabilities do not differentiate between the final etching products. For example, the etching products related to Γ_{SiF_3} can be either SiF_x or Si_xN_y .

We determine the partial etching yields and the reaction probabilities defined above by analyzing our MD simulations during the steady-state period (100–200 HF/nm^2 in Figure 6a). Parameters are computed within the steady-state region because the second assumption of the model posits that the probabilities are independent of the surface states. The Si and N atoms within the modified layer are labeled as SiF_n ($n = 0, 1, 2, 3$) and NH_m ($m = 0, 1, 2$), respectively, as defined in eqs 8 and 9. For each HF incidence, we trace the changes among these surface species of the interaction volume. We then

enumerate the reaction and etching (desorption) events across the entire slab, which can occur beneath the volume due to the penetration of the energetic ion.

The conversions involving multiple ligand changes (e.g., from SiF_0 to SiF_2) are treated as a series of single steps (e.g., from SiF_0 to SiF_1 and from SiF_1 to SiF_2), increasing counts for the two processes. When the number of attached H and F atoms decrease (e.g., from SiF_2 to SiF_1), the counts are reduced. By normalizing the reaction and desorption counts with the number of corresponding reactants and dose (100 HF/nm^2), we obtain reaction and desorption probabilities for the given HF incidence. By averaging these values over the whole HF incidences during the steady state, we calculate the coarse-grained parameters r_n , q_m , Γ_{SiF_n} , and Γ_{NH_m} , which are presented in Table S4. It is worth noting that the coarse-grained parameters for different combinations of E_{in} and θ_{in} can be computed in a similar way, provided MD simulations are conducted. For example, the energy dependence of Γ_i when $\theta_{in} = 0^\circ$ is shown in Figure S6. A full parametrization of the energy and angular dependence would be done in future.

Using the parameters obtained in the above, we construct a mesoscale 1D continuum model that produces etching yields as well as the evolution of surface states. We first define N_{SiF_n} and N_{NH_m} as the surface densities of corresponding surface species within the modified layer. We assume that the etched Si and N atoms are compensated by additional SiF_0 and NH_0 from the bulk region. That is to say, the number of Si atoms within the modified layer remains constant. Thus

$$\partial_t N_{\text{SiF}_0} = (-r_0 N_{\text{SiF}_0} - \Gamma_{\text{SiF}_0} N_{\text{SiF}_0} + Y_{\text{Si}}) J_{\text{HF}} \quad (10)$$

$$\partial_t N_{\text{SiF}_1} = (r_0 N_{\text{SiF}_0} - r_1 N_{\text{SiF}_1} - \Gamma_{\text{SiF}_1} N_{\text{SiF}_1}) J_{\text{HF}} \quad (11)$$

$$\partial_t N_{\text{SiF}_2} = (r_1 N_{\text{SiF}_1} - r_2 N_{\text{SiF}_2} - \Gamma_{\text{SiF}_2} N_{\text{SiF}_2}) J_{\text{HF}} \quad (12)$$

$$\partial_t N_{\text{SiF}_3} = (r_2 N_{\text{SiF}_2} - \Gamma_{\text{SiF}_3} N_{\text{SiF}_3}) J_{\text{HF}} \quad (13)$$

$$\partial_t N_{\text{NH}_0} = \left(-q_0 N_{\text{NH}_0} - \Gamma_{\text{NH}_0} N_{\text{NH}_0} + \frac{4}{3} Y_{\text{Si}} \right) J_{\text{HF}} \quad (14)$$

$$\partial_t N_{\text{NH}_1} = (q_0 N_{\text{NH}_0} - q_1 N_{\text{NH}_1} - \Gamma_{\text{NH}_1} N_{\text{NH}_1}) J_{\text{HF}} \quad (15)$$

$$\partial_t N_{\text{NH}_2} = (q_1 N_{\text{NH}_1} - \Gamma_{\text{NH}_2} N_{\text{NH}_2}) J_{\text{HF}} \quad (16)$$

where J_{HF} is the flux of incoming HF and $Y_{\text{Si}} = \sum_{n=0}^3 \Gamma_{\text{SiF}_n} N_{\text{SiF}_n}$ is the total etching yield of Si. In eq 14, N atoms are supplied in proportion to the supply of Si atoms, based on the stoichiometric ratio of the bulk region. We note that the time scale of this model can be extrapolated by substituting the ion

flux in eqs 10–16 with experimental values. However, the model does not account for long-term reactions (see Section 3.2).

We numerically solve the above set of simultaneous differential equations using MATLAB.⁷⁰ The computed numbers of surface species (N_i) from the continuum model are plotted in Figure 12. For comparison, the counts from the MD simulations are also presented. Populations of surface species in Figure 12a,b are averaged over three different MD simulations for visual clarity, while the ratio of species in Figure 12c is not averaged to show the distribution of individual MD results. Although the parameters in the continuum model were derived based on the steady-state results, the mesoscale model can still describe the initial stage reasonably. This supports that the model well captures the underlying reaction mechanism.

In Figure 12a, $N_{\text{Si}} = \sum_{n=0}^3 N_{\text{SiF}_n}$ from the continuum model and MD simulations remains constant while there is a noticeable increase in Si fluorination as the etching process advances. In Figure 12b, the total number of N atoms declines at the onset of the etching process. This is attributed to the higher etching yields of NH_0 and NH_1 compared to SiF_0 and SiF_1 (see Table S4), contributed by the formation of N_2 . However, as the etching process continues, the etching yield of Si increases with more fluorinated Si species, halting the further depletion of N. Figure 12c shows the ratio of N to Si. Compared to the bulk stoichiometry of 1.33 (indicated by a horizontal dashed line), the initial decrease related to N depletion, as discussed in Section 3.2.1, is well reproduced in the present model.

We have successfully demonstrated the integration of detailed microscale simulation data into a mesoscale model. This bottom-up approach to scale bridging would be useful for accurately predicting the feature profiles in plasma etching, which are significantly influenced by plasma–surface interactions.⁷ This methodology would be particularly effective because it allows for the dynamic representation of surface composition changes. As a result, it enhances the modeling precision for advanced plasma etching techniques, such as time-multiplexed processes where the surface composition varies actively over time.^{4,5}

4. CONCLUSION

In summary, we have developed an accurate NNP designed for dry etching of a- Si_3N_4 using HF molecules. The surface reactions in diverse local environments were sampled by incorporating several types of training set; baseline structures, reaction-specific data, together with general-purpose training sets. Furthermore, the NNP was refined through iterative comparisons with DFT results. Using the trained NNP, we conducted etching simulations with various kinetic energies and incident angles of HF molecules. The etching simulation allowed for detailed observation and analysis of key processes such as preferential sputtering, surface modification, etching yield, threshold energy, and the distribution of etching products. Furthermore, we identified and characterized the formation of a modified surface layer, elucidating its role in chemical sputtering. The observed surface modifications are primarily attributed to preferential sputtering, which involves the generation of N_2 molecules. We also revealed the angle dependence of the etching mechanism and yield, influenced by variations in reflection coefficients and the development of the modified layer. Finally, we developed a simple mesoscale

model, built from our MD simulation results, which effectively reproduces the surface composition obtained with MD simulations. By establishing a computational framework for atomistic etching simulation and scale bridging, this work paves the way for a more accurate and efficient design of etching processes in the semiconductor industry, thereby enhancing device performance and manufacturing precision.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c07949>.

Supporting Information: this file includes cutoff distances of the switch function; parameters used in the symmetry functions; snapshots of a high-temperature MD simulation; parity plots of energy, force component, and stress component, radial and angular distribution functions of amorphous Si_3N_4 ; etching yields of H-terminated and pristine a- Si_3N_4 ; surface profiles of H-terminated and pristine a- Si_3N_4 substrates after etching simulations; atomic density profiles at surfaces under different E_{in} ; atomic density profiles of surfaces with different areas; desorption and reaction probabilities used in the coarse-grained model; desorption probabilities of Si surface species as a function of E_{in} (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Seungwu Han – Department of Materials Science and Engineering and Research Institute of Advanced Materials, Seoul National University, Seoul 08826, Republic of Korea; Korea Institute for Advanced Study, Seoul 02455, Republic of Korea; orcid.org/0000-0003-3958-0922; Phone: +82 (02)880 1541; Email: hansw@snu.ac.kr

Authors

Changho Hong – Department of Materials Science and Engineering and Research Institute of Advanced Materials, Seoul National University, Seoul 08826, Republic of Korea; orcid.org/0000-0001-8163-5443

Sangmin Oh – Department of Materials Science and Engineering and Research Institute of Advanced Materials, Seoul National University, Seoul 08826, Republic of Korea; orcid.org/0000-0002-4229-3302

Hyungmin An – Department of Materials Science and Engineering and Research Institute of Advanced Materials, Seoul National University, Seoul 08826, Republic of Korea; orcid.org/0000-0003-2529-4247

Purun-hanul Kim – Department of Materials Science and Engineering and Research Institute of Advanced Materials, Seoul National University, Seoul 08826, Republic of Korea; orcid.org/0000-0002-6338-7957

Yaeji Kim – SK Hynix Inc, Icheon-si, Gyeonggi-do 17336, Republic of Korea

Jae-hyeon Ko – SK Hynix Inc, Icheon-si, Gyeonggi-do 17336, Republic of Korea

Jiwoong Sue – SK Hynix Inc, Icheon-si, Gyeonggi-do 17336, Republic of Korea

Dongyeon Oh – SK Hynix Inc, Icheon-si, Gyeonggi-do 17336, Republic of Korea; orcid.org/0000-0002-1262-1838

Sungkye Park — SK Hynix Inc, Icheon-si, Gyeonggi-do 17336, Republic of Korea

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acsami.4c07949>

Author Contributions

^{||}C.H. and S.O. equally contributed to this work.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank Yutack Park for accelerating SIMPLE-NN MD code. This paper was the result of the research project supported by SK hynix. The authors acknowledge the financial support from the Virtual Engineering Platform Project of the Ministry of Trade, Industry and Energy (MOTIE) of Korea (grant number: P0022336-G02F09901883413-10054408). The computations were carried out at Korea Institute of Science and Technology Information (KISTI) National Supercomputing Center (KSC-2023-CRE-0458) and at the Center for Advanced Computations (CAC) at Korea Institute for Advanced Study (KIAS).

REFERENCES

- (1) Shen, M.; Lill, T.; Hoang, J.; Chi, H.; Routzahn, A.; Church, J.; Subramonium, P.; Puthenkovilakam, R.; Reddy, S.; Bhadauriya, S.; Roberts, S.; Kamarthy, G. Progress report on high aspect ratio patterning for memory devices. *Jpn. J. Appl. Phys.* **2023**, *62*, S1801.
- (2) Kihara, Y.; Tomura, M.; Sakamoto, W.; Honda, M.; Kojima, M. Beyond 10 μm depth ultra-high speed etch process with 84% lower carbon footprint for memory channel hole of 3D NAND flash over 400 layers. In *2023 IEEE Symposium on VLSI Technology and Circuits*, 2023; pp 1–2.
- (3) Schaepkens, M.; Standaert, T. E. F. M.; Rueger, N. R.; Sebel, P. G. M.; Oehrlein, G. S.; Cook, J. M. Study of the SiO_2 -to- Si_3N_4 etch selectivity mechanism in inductively coupled fluorocarbon plasmas and a comparison with the SiO_2 -to-Si mechanism. *J. Vac. Sci. Technol. A* **1999**, *17*, 26–37.
- (4) Dallorto, S.; Goodyear, A.; Cooke, M.; Szornel, J. E.; Ward, C.; Kastl, C.; Schwartzberg, A.; Rangelow, I. W.; Cabrini, S. Atomic layer etching of SiO_2 with Ar and CHF_3 plasmas: a self-limiting process for aspect ratio independent etching. *Plasma Processes Polym.* **2019**, *16*, 1900051.
- (5) Yu, A.; Lau, J. H.; Ho, S. W.; Kumar, A.; Hnin, W. Y.; Lee, W. S.; Jong, M. C.; Sekhar, V. N.; Kripesh, V.; Pinjala, D.; et al. Fabrication of high aspect ratio TSV and assembly with fine-pitch low-cost solder microbump for Si interposer technology with high-density interconnects. In *IEEE Transactions on Components, Packaging and Manufacturing Technology*, 2011, pp 1336–1344.
- (6) Liu, Z.; Wu, Y.; Harteneck, B.; Olynick, D. Super-selective cryogenic etching for sub-10 nm features. *Nanotechnol.* **2013**, *24*, 015305.
- (7) Marchack, N.; Chang, J. P. Perspectives in nanoscale plasma etching: what are the ultimate limits? *J. Phys. D: Appl. Phys.* **2011**, *44*, 174011.
- (8) Vanraes, P.; Parayil Venugopalan, S.; Bogaerts, A. Multiscale modeling of plasma–surface interaction—general picture and a case study of Si and SiO_2 etching by fluorocarbon-based plasmas. *Appl. Phys. Rev.* **2021**, *8*, 041305.
- (9) Kline, L. E.; Kushner, M. J. Computer simulation of materials processing plasma discharges. *Crit. Rev. Solid State Mater. Sci.* **1989**, *16*, 1–35.
- (10) Kim, H. C.; Manousiouthakis, V. I. Simulation based plasma reactor design for improved ion bombardment uniformity. *J. Vac. Sci. Technol. B* **2000**, *18*, 841–847.
- (11) Guo, W.; Sawin, H. H. Review of profile and roughening simulation in microelectronics plasma etching. *J. Phys. D: Appl. Phys.* **2009**, *42*, 194014.
- (12) Hoekstra, R. J.; Grapperhaus, M. J.; Kushner, M. J. Integrated plasma equipment model for polysilicon etch profiles in an inductively coupled plasma reactor with subwafer and superwafer topography. *J. Vac. Sci. Technol. A* **1997**, *15*, 1913–1921.
- (13) Yook, Y. G.; You, H. S.; Park, J. H.; Chang, W. S.; Kwon, D. C.; Yoon, J. S.; Yoon, K. H.; Shin, S. S.; Yu, D. H.; Im, Y. H. Fast and realistic 3D feature profile simulation platform for plasma etching process. *J. Phys. D: Appl. Phys.* **2022**, *55*, 255202.
- (14) Neyts, E. C.; Brault, P. Molecular dynamics simulations for plasma-surface interactions. *Plasma Processes Polym.* **2017**, *14*, 1600145.
- (15) Abrams, C. F.; Graves, D. B. Molecular dynamics simulations of Si etching by energetic CF_3^+ . *J. Appl. Phys.* **1999**, *86*, 5938–5948.
- (16) Tinacba, E. J. C.; Ito, T.; Karahashi, K.; Isobe, M.; Hamaguchi, S. Molecular dynamics simulation for reactive ion etching of Si and SiO_2 by SF_5^+ ions. *J. Vac. Sci. Technol. B* **2021**, *39*, 043203.
- (17) Kim, D. H.; Kwak, S. J.; Jeong, J. H.; Yoo, S.; Nam, S. K.; Kim, Y.; Lee, W. B. Molecular dynamics simulation of silicon dioxide etching by hydrogen fluoride using the reactive force field. *ACS Omega* **2021**, *6*, 16009–16015.
- (18) Xu, J.; Lu, K.; Fan, D.; Wang, Y.; Xu, S.; Kubo, M. Different etching mechanisms of diamond by oxygen and hydrogen plasma: a reactive molecular dynamics study. *J. Phys. Chem. C* **2021**, *125*, 16711–16718.
- (19) Diddens, D.; Appiah, W. A.; Mabrouk, Y.; Heuer, A.; Vegge, T.; Bhowmik, A. Modeling the solid electrolyte interphase: machine learning as a game changer? *Adv. Mater. Interfaces* **2022**, *9*, 2101734.
- (20) Watanabe, S.; Li, W.; Jeong, W.; Lee, D.; Shimizu, K.; Mimanitani, E.; Ando, Y.; Han, S. High-dimensional neural network atomic potentials for examining energy materials: some recent simulations. *J. Phys.: Energy* **2021**, *3*, 012003.
- (21) Hong, C.; Choi, J. M.; Jeong, W.; Kang, S.; Ju, S.; Lee, K.; Jung, J.; Youn, Y.; Han, S. Training machine-learning potentials for crystal structure prediction using disordered structures. *Phys. Rev. B* **2020**, *102*, 224104.
- (22) Lee, K.; Park, Y.; Han, S. *Ab initio* construction of full phase diagram of MgO–CaO eutectic system using neural network interatomic potentials. *Phys. Rev. Mater.* **2022**, *6*, 113802.
- (23) Choi, J. M.; Lee, K.; Kim, S.; Moon, M.; Jeong, W.; Han, S. Accelerated computation of lattice thermal conductivity using neural network interatomic potentials. *Comput. Mater. Sci.* **2022**, *211*, 111472.
- (24) Wang, C.; Aoyagi, K.; Wisesa, P.; Mueller, T. Lithium ion conduction in cathode coating materials from on-the-fly machine learning. *Chem. Mater.* **2020**, *32*, 3741–3752.
- (25) Chen, D.; Shang, C.; Liu, Z.-P. Machine-learning atomic simulation for heterogeneous catalysis. *npj Comput. Mater.* **2023**, *9*, 2.
- (26) Chen, L.; Sukuba, I.; Probst, M.; Kaiser, A. Iterative training set refinement enables reactive molecular dynamics via machine learned forces. *RSC Adv.* **2020**, *10*, 4293–4299.
- (27) Xia, Y.; Sautet, P. Plasma oxidation of copper: molecular dynamics study with neural network potentials. *ACS Nano* **2022**, *16*, 20680–20692.
- (28) Hong, C.; Kim, J.; Kim, J.; Jung, J.; Ju, S.; Choi, J. M.; Han, S. Applications and training sets of machine learning potentials. *Sci. Technol. Adv. Mater.: Methods* **2023**, *3*, 2269948.
- (29) Donnelly, V. M.; Kornblit, A. Plasma etching: yesterday, today, and tomorrow. *J. Vac. Sci. Technol. A* **2013**, *31*, 050825.
- (30) Hirata, A.; Fukasawa, M.; Kugimiya, K.; Nagaoka, K.; Karahashi, K.; Hamaguchi, S.; Iwamoto, H. On-wafer monitoring and control of ion energy distribution for damage minimization in atomic layer etching processes. *Jpn. J. Appl. Phys.* **2020**, *59*, SJJC01.
- (31) Wood, M. A.; Cusentino, M. A.; Wirth, B. D.; Thompson, A. P. Data-driven material models for atomistic simulation. *Phys. Rev. B* **2019**, *99*, 184305.

- (32) Wang, Y.; Liu, J.; Li, J.; Mei, J.; Li, Z.; Lai, W.; Xue, F. Machine-learning interatomic potential for radiation damage effects in bcc-iron. *Comput. Mater. Sci.* **2022**, *202*, 110960.
- (33) Byggmästar, J.; Hamedani, A.; Nordlund, K.; Djurabekova, F. Machine-learning interatomic potential for radiation damage and defects in tungsten. *Phys. Rev. B* **2019**, *100*, 144105.
- (34) Biersack, J. P.; Ziegler, J. F. *The Stopping and Range of Ions in Solids*; Ion implantation techniques: Berlin, Heidelberg, 1982; pp 122–156.
- (35) Sandia Corporation. *LAMMPS Documentation* (7 Feb 2024 Version); Sandia Corporation, 2024 <https://docs.lammps.org/Manual.html> (accessed 2024–02–29).
- (36) Lee, K.; Yoo, D.; Jeong, W.; Han, S. SIMPLE-NN: an efficient package for training and executing neural-network interatomic potentials. *Comput. Phys. Commun.* **2019**, *242*, 95–103.
- (37) Kingma, D. P.; Ba, J. Adam: a method for stochastic optimization. *arXiv* **2014**, arXiv:1412.6980v9.
- (38) Kresse, G.; Furthmüller, J. Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set. *Comput. Mater. Sci.* **1996**, *6*, 15–50.
- (39) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **1996**, *77*, 3865–3868.
- (40) Hoover, W. G. Canonical dynamics: equilibrium phase-space distributions. *Phys. Rev. A* **1985**, *31*, 1695–1697.
- (41) Plimpton, S. Fast parallel algorithms for short-range molecular dynamics. *J. Comput. Phys.* **1995**, *117*, 1–19.
- (42) Nordlund, K. Molecular dynamics simulation of ion ranges in the 1–100 keV energy range. *Comput. Mater. Sci.* **1995**, *3*, 448–456.
- (43) Grimme, S. Semiempirical GGA-type density functional constructed with a long-range dispersion correction. *J. Comput. Chem.* **2006**, *27*, 1787–1799.
- (44) Yen, B. K.; White, R. L.; Waltman, R. J.; Dai, Q.; Miller, D. C.; Kellock, A. J.; Marchon, B.; Kasai, P. H.; Toney, M. F.; York, B. R.; Deng, H.; Xiao, Q.-F.; Raman, V. Microstructure and properties of ultrathin amorphous silicon nitride protective coating. *J. Vac. Sci. Technol. A* **2003**, *21*, 1895–1904.
- (45) Sheft, I.; Perkins, A. J.; Hyman, H. H. Anhydrous hydrogen fluoride: vapor pressure and liquid density. *J. Inorg. Nucl. Chem.* **1973**, *35*, 3677–3680.
- (46) Ryckaert, J.-P.; Ciccotti, G.; Berendsen, H. J. Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes. *J. Comput. Phys.* **1977**, *23*, 327–341.
- (47) Carter, E.; Ciccotti, G.; Hynes, J. T.; Kapral, R. Constrained reaction coordinate dynamics for the simulation of rare events. *Chem. Phys. Lett.* **1989**, *156*, 472–477.
- (48) Liu, L.-H.; Michalak, D. J.; Chopra, T. P.; Pujari, S. P.; Cabrera, W.; Dick, D.; Veyan, J.-F.; Hourani, R.; Halls, M. D.; Zuillhof, H.; Chabal, Y. J. Surface etching, chemical modification and characterization of silicon nitride and silicon oxide—selective functionalization of Si₃N₄ and SiO₂. *J. Phys.: Condens. Matter* **2016**, *28*, 094014.
- (49) Tatsumi, T.; Hikosaka, Y.; Morishita, S.; Matsui, M.; Sekine, M. Etch rate control in a 27 MHz reactive ion etching system for ultralarge scale integrated circuit processing. *J. Vac. Sci. Technol. A* **1999**, *17*, 1562–1569.
- (50) Cagomoc, C. M. D.; Isobe, M.; Hudson, E. A.; Hamaguchi, S. Molecular dynamics simulation of oxide-nitride bilayer etching with energetic fluorocarbon ions. *J. Vac. Sci. Technol. A* **2022**, *40*, 063006.
- (51) Gotoh, Y.; Kure, T.; Tachi, S. Estimation of ion incident angle from Si etching profiles. *Jpn. J. Appl. Phys.* **1993**, *32*, 3035.
- (52) Steinbrüchel, C. Universal energy dependence of physical and ion-enhanced chemical etch yields at low ion energy. *Appl. Phys. Lett.* **1989**, *55*, 1960–1962.
- (53) Yanai, K.-i.; Karahashi, K.; Ishikawa, K.; Nakamura, M. Mass-analyzed CF_x⁺ (x = 1,2,3) ion beam study on selectivity of SiO₂-to-SiN etching and a-C:F film deposition. *J. Appl. Phys.* **2005**, *97*, 053302.
- (54) Kastenmeier, B. E. E.; Matsuo, P. J.; Beulens, J. J.; Oehrlein, G. S. Chemical dry etching of silicon nitride and silicon dioxide using CF₄/O₂/N₂ gas mixtures. *J. Vac. Sci. Technol. A* **1996**, *14*, 2802–2813.
- (55) Kastenmeier, B. E. E.; Matsuo, P. J.; Oehrlein, G. S.; Langan, J. G. Remote plasma etching of silicon nitride and silicon dioxide using NF₃/O₂ gas mixtures. *J. Vac. Sci. Technol. A* **1998**, *16*, 2047–2056.
- (56) Clarke, P. E.; Field, D.; Hydes, A. J.; Klemperer, D. F.; Seakins, M. J. Mass spectrometric studies of plasma etching of silicon nitride. *J. Vac. Sci. Technol. B* **1985**, *3*, 1614–1619.
- (57) Field, D.; Klemperer, D. F.; Wade, I. T. Spectroscopic studies of fluorescent emission in plasma etching of silicon nitride. *J. Vac. Sci. Technol. B* **1988**, *6*, 551–558.
- (58) Winters, H. F. Etch products from the reaction of XeF₂ with SiO₂, Si₃N₄, SiC, and Si in the presence of ion bombardment. *J. Vac. Sci. Technol. B* **1983**, *1*, 927–931.
- (59) Zhou, N.-S.; Fujita, S.; Sasaki, A. Structural and electrical properties of plasma-deposited silicon nitride from SiH₄-N₂ gas mixture. *J. Electron. Mater.* **1985**, *14*, 55–72.
- (60) Stein, H. J.; Wells, V. A.; Hampy, R. E. Properties of plasma-deposited silicon nitride. *J. Electrochem. Soc.* **1979**, *126*, 1750–1754.
- (61) Kuboi, N.; Tatsumi, T.; Kinoshita, T.; Shigetoshi, T.; Fukasawa, M.; Komachi, J.; Ansai, H. Prediction of plasma-induced damage distribution during silicon nitride etching using advanced three-dimensional voxel model. *J. Vac. Sci. Technol. A* **2015**, *33*, 061308.
- (62) Dulak, J.; Howard, B. J.; Steinbrüchel, C. Etch mechanism in the reactive ion etching of silicon nitride. *J. Vac. Sci. Technol. A* **1991**, *9*, 775–778.
- (63) Timms, P. L.; Kent, R. A.; Ehlert, T. C.; Margrave, J. L. Silicon-fluorine chemistry. I. silicon difluoride and the perfluorosilanes. *J. Am. Chem. Soc.* **1965**, *87*, 2824–2828.
- (64) Gray, D. C.; Tepermeister, I.; Sawin, H. H. Phenomenological modeling of ion-enhanced surface kinetics in fluorine-based plasma etching. *J. Vac. Sci. Technol. B* **1993**, *11*, 1243–1257.
- (65) Eckstein, W.; Biersack, J. P. Reflection of heavy ions. *Z. Phys. B: Condens. Matter* **1986**, *63*, 471–478.
- (66) Langley, R. A.; Bohdansky, J.; Eckstein, W.; Mioduszewski, P.; Roth, J.; Taglauer, E.; Thomas, E. W.; Verbeek, H.; Wilson, K. L. Data compendium for plasma-surface interactions. *Nucl. Fusion* **1984**, *24*, S9.
- (67) Hamblen, D. P.; Cha-Lin, A. Angular etching correlations from RIE: application to VLSI fabrication and process modeling. *J. Electrochem. Soc.* **1988**, *135*, 1816–1822.
- (68) Grill, V.; Shen, J.; Evans, C.; Cooks, R. G. Collisions of ions with surfaces at chemically relevant energies: instrumentation and phenomena. *Rev. Sci. Instrum.* **2001**, *72*, 3149–3179.
- (69) Lee, J.-K.; Lee, G.-R.; Min, J.-H.; Moon, S. H. Angular dependence of Si₃N₄ etch rates and the etch selectivity of SiO₂ to Si₃N₄ at different bias voltages in a high-density C₄F₈ plasma. *J. Vac. Sci. Technol. A* **2007**, *25*, 1395–1401.
- (70) *MATLAB Release R2023a*; The MathWorks Inc: Natick, Massachusetts, United States, 2023 <https://www.mathworks.com/products/matlab> (accessed 2024–02–29).